Python Basics - Lab #3 of 3

This is the third lab in a three-part series that covers Python fundamentals. In this lab, we will focus on building loops using FOR and WHILE, along with creating and executing a Python program.

Creating Loops with FOR and WHILE

In this section, we are going to look at looping. Looping is powerful within a programming language.

1. Let's begin with the following example, where we will recreate the inventory list: `inventory = ["Catalyst 3750", "Catalyst 2960", "2911 Router", "7965G IP Phone", "CUCME Server"]`. We will be working in the interactive interpreter here. Plug in these commands below:
    - inventory = ["Catalyst 3750", "Catalyst 2960", "2911 Router", "7965G IP Phone", "CUCME Server"]
    - for item in inventory:
    - print(item)
        - *(What this will do is examine the first item in the inventory list and assign it to the variable `item`. It will then execute any indented code. Afterward, it will move to the second item in the list and repeat the process, executing the indented code for each subsequent item. It will keep looping until there are no more items in the list.)*
        - *(Make sure to indent your print command then hit **Enter** twice.)*

```
>>> inventory = ["Catalyst 3750", "Catalyst 2960", "2911 Router", "7965G IP Phone", "CUCME Server"]
>>> for item in inventory:
...     print(item)
...
Catalyst 3750
Catalyst 2960
2911 Router
7965G IP Phone
CUCME Server
>>>
```

2. Next, we'll focus on the WHILE loop, which repeatedly executes a block of code until a specified condition is met. In the example below, we'll prompt the user for their name. If they don't provide it, the program will keep asking until a name is provided. This will be done in the IDLE environment.
    - name=input("Enter your name: ")
    - while name=="":
        - *(What we are saying here is while this condition is true, where a name wasn't inputted, continue to execute the two lines of code below.)*
    - print("Please enter your name!!!")
    - name=input("Enter your name:")print(f"Hello {name}")

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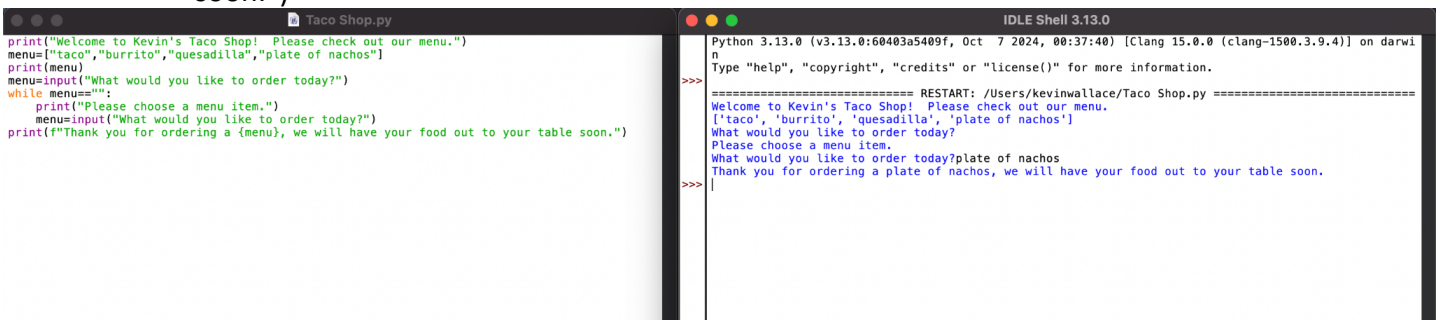
- *(Once a name is entered, the while loop will terminate, and "Hello 'Name'" will be displayed.)*
- *(Now, save the file and run it. When prompted for your name, enter it as requested.)*
- *(The f-string tells the program to fill the variable of `name` with our name whenever we type it in and press enter.)*

Creating and Executing a Python Program

In this final section, we are going to have some fun with creating a program and incorporating some of the things that we have learned throughout this series.

1. We are going to create a taco shop and incorporate some of the lessons that we have learned throughout this series. We will be using the IDLE environment in this section.
    - print("Welcome to 'First Name' Taco Shop! Please check out our menu.")
    - menu=["taco","burrito","quesadilla","plate of nachos"]
    - print(menu)
    - menu=input("What would you like to order today?")
    - while menu=="":
    - print("Please choose a menu item.")
    - menu=input("What would you like to order today?")
    - print(f"Thank you for ordering a {menu}, we will have your food out to your table soon.")



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